Engineering Metric Prefixes – Quick Reference

Metric prefixes represent very large or very small numbers using powers of ten. They make values easier to read, compare, and calculate.

Common Large Prefixes

kilo	k	10^3	thousand
mega	M	10^6	million
giga	G	10^9	billion
tera	T	10^{12}	trillion
peta	P	10^{15}	Quadrillion

Common Small Prefixes

milli	m	10^{-3}	one-thousandth
micro	μ	10^{-6}	one-millionth
nano	n	10^{-9}	one-billionth
pico	p	10^{-12}	one-trillionth

Engineering Notation:

- Uses steps of **10^3** (thousands).
- Keeps numbers between **1 and 999** for readability.
- Example: $47{,}000\,\Omega \;=\; 47\,k\Omega$

Metric Conversions

Metric conversion is changing a value from one metric unit to another by shifting the decimal point according to powers of ten.

Example: **larger → smaller unit**

- $5\,k\Omega \;=\; 5 \times 10^3\,\Omega \;=\; 5{,}000\,\Omega$
- $2.5\,M\Omega = 2.5 \times 10^6\,\Omega = 2{,}500{,}000\,\Omega$
- $3.6\,G\Omega = 3.6 \times 10^9\,\Omega$

Example: **Smaller → Larger unit**

- $12{,}000\,\Omega = 12 \times 10^3 = \;12\,k\Omega$
- $0.00047\,F = 470 \times 10^{-6} = 470\,\mu F$
- $0.00000025\,s \;= 250 \times 10^{-9} = 250ns$

Metric Arithmetic:

Always convert to the **same prefix** before adding or subtracting:

e.g., $10{,}000\,\Omega + 22\,k\Omega = \;\; 10{,}000\Omega + 22{,}000\Omega \;=\; 32{,}000\Omega$ or $10k\Omega + 22k\Omega = 32k\Omega$